

MineGDS™:

Designing Microchips, One Block at a Time

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Abstract—Integrated circuit design has problems. We have fixed them. Our approach takes cutting-edge technology and merges it with the fun the discipline has been missing. Microchip design will never be the same. Your reality is about to change. You will be moved, so prepare your tissues. After the tears, it will be like seeing color for the first time. Prepare your wallets; no other product will make you reach for them faster. The future is here, and it is arriving at blazing speeds. Are you ready for it?

Index Terms—integrated circuits, EDA, hardware, minecraft, gamification, revolutionary, cow

I. INTRODUCTION

The field of integrated circuit (IC) design has rapidly accelerated over the past few decades, with at least 12 ICs [1] being produced annually. Our market research has shown that electronic design automation (EDA) tools for IC design lacks the accessibility and availability of equivalent software EDA tools. This may therefore form a significant barrier to those wanting to enter the field of IC design.

At the time of writing (2026 AD, around lunchtime), the IC EDA landscape is predominantly divided among a small number of established tools, incl. the software suites provided by *Cadence* and *Synopsis* [2]. These tools are required for designing IC layouts in the GDSII file format, which can then be manufactured at a semiconductor foundry. These tools typically require specialist training for effective use, and the model files required for designing an IC at one of these foundries are often closed-source.

Over the past few years, a number of these foundries have started to offer open-source model files, including those by *SkyWater* and *GlobalFoundries* [3], [4]. This enables IC design using fully open-source software and may open up the field to designers outside of large enterprise settings.

However, while these open-source tools are free to access and use, their interfaces (and interfacing with multiple programs) leaves room for further refinement. The user interfaces of established tools like *XSchem* and *KLayout* allow for experienced designers to quickly get used to this new workflow, but may still offer a steep learning curve to those who are new to IC design.

To this end, MOST, Inc. ®™© (*Masters Of Semiconductor Technology, Inc.*) has developed a novel tool for the design of GDSII files, required for producing one's own IC at a foundry. You see, as children, the authors of this contribution yearned

for the mines [5], but something always got in the way. Despite that, the call of the mines was too strong. So one day, they started digging.

And digging.

And digging.

Until they found it. The user interface for editing GDSII-files they had always been waiting for. Tears rolled down the authors' cheeks while basking in the glory of the product they had developed. Could it really have been this beautiful, elegant and simple all along? Could IC design finally become a field for the masses? Would this introduction ask more rhetorical questions?

Introducing MineGDS™.

II. METHODS

A. User Experience

MOST, Inc. ®™© (*Marine Otter Security Team, Inc.*) MineGDS™ offers a user-friendly interface for editing GDSII-files. Simply provide a GDSII-file of choice and MineGDS™ will export it to a Minecraft [6] world. After *editing* this world by any means the user desires, this world can be converted back to a GDSII-file using MineGDS™. The trademark office told us we can only protect our trademark by using it as much as possible. Thus: MineGDS™, MineGDS™, MineGDS™.

B. Technical Description¹

Basically, you use a Python library to convert each layer in a GDSII file to a PNG image file, after which you use this other Python library that lets you edit blocks on a Minecraft [6] server to place the blocks. Then you can do whatever you want in Minecraft [6]. Finally, that same Python library can read out the blocks again and put them into a PNG. This PNG then needs to be converted back to GDSII using that same other Python library. However, first we optimized the rasterization of the blocks into larger rectangles by vibe coding an algorithm, because we didn't want to get like 10 million separate blocks as rectangles in the GDSII-file. TL;DR: Vibe code some shit and discover that Java doesn't support emoji which sends you down the rabbit hole of debugging for a week (thank you ChatGPT, not for debugging, but for ruining my career.), then convince your bosses that it took you at least a year to get this working (I love under-promising).

¹ Kindly provided by our extremely over-cafeinated I(an) T. Guy.



Fig. 1: Example use case of MineGDS™: A reference NAND gate design is *edited* using TNT in MineGDS™ to obtain a (slightly damaged) AND gate.

C. Maximizing Profit Value Proposition

Think of the immense value that MineGDS™ provides to its users. There's the conversion from GDSII to Minecraft [6]. And, as if that wasn't enough, it also provides conversion **back** to GDSII! Considering the telemetry services that the maintenance of such a wonderful tool requires, MineGDS™ will launch at the low, low price of \$19,99/month.

Naturally, if you would like to include design rule checking (DRC) to ensure that an exported design is "manufacturable" at a foundry of choice, that can be provided too. For a small surcharge of \$219,99/month your designs can be verifiably "manufacturable", synergetically aligned and coincidentally serendipitous [7].

D. Test Setup¹

To demonstrate the abilities of MineGDS™, we will show the editing of a NAND gate layout that we found somewhere [8]. We want to turn the NAND gate into an AND gate. Surely that means that you have to remove the inverting wire that is somewhere in the layout, therefore we shall use TNT inside MineGDS™. Finally we hope to end up with a functional AND gate. Fig. 1 gives a schematic overview of the test setup.

III. RESULTS

A. Example Usage¹

Fig. 2 shows in-game screenshots of MineGDS™, loaded in Minecraft [6]. In Fig. 2a you can see that we have loaded the layout of the NAND gate. Each layer of the GDSII-file is represented by a configurable layer of blocks, in this case we use colored wool because we wanted to get comfy and it reminds me of the knitted blankets my grandma made for me. After this we show the editing of the layout in MineGDS™. Since we simply want to remove a wire, the MOST efficient way of doing this in Minecraft [6] is by using TNT [9]. This process is show in Fig. 2b. The final results of the editing in MineGDS™ are shown in Fig. 2c. To prove that this actually works we also show a converted layout from MineGDS™ in KLayout in Fig. 3, which obviously would be a lot of work to create manually (Trust me bro) (no but this actually works, no joke). It is notable that an actual Minecraft [6] village has appeared in the chip layout. I'm pretty sure this would be referred to as "emergent properties".

B. Financial Results

MOST, Inc. ®™© (May Overcomplicate Several Trivialities, Inc.) has opted to keep this strategic information confidential. Just know that money buys happiness [10] and that we



(a) NAND layout loaded into MineGDS™.



(b) Editing in MineGDS™. It is as easy as placing TNT to annihilate the surrounding blocks. Be careful to not be close to the TNT once it goes off.



(c) Finished editing?

Fig. 2: Screen captures of usage of MineGDS™.

are very happy at MOST, Inc. ®™© (Money Often Simply Terminates, Inc.). - Steve Qubic². Instead, have a highly compressed image of "Steve". With such a groundbreaking product we will obviously trade publicly. The values of our stocks will be proportional to the number of errors in your circuit design.

$$\text{MOST}_{\text{share}} \propto (\# \text{ MineGDS users}) \cdot (\# \text{ IC design errors}) \quad (1a)$$

If you look at equation (1a) and consider that there are 1bn stocks of MOST, Inc. ®™© (*Massively Overvalued Stock*)

²This is not an admission of guilt.

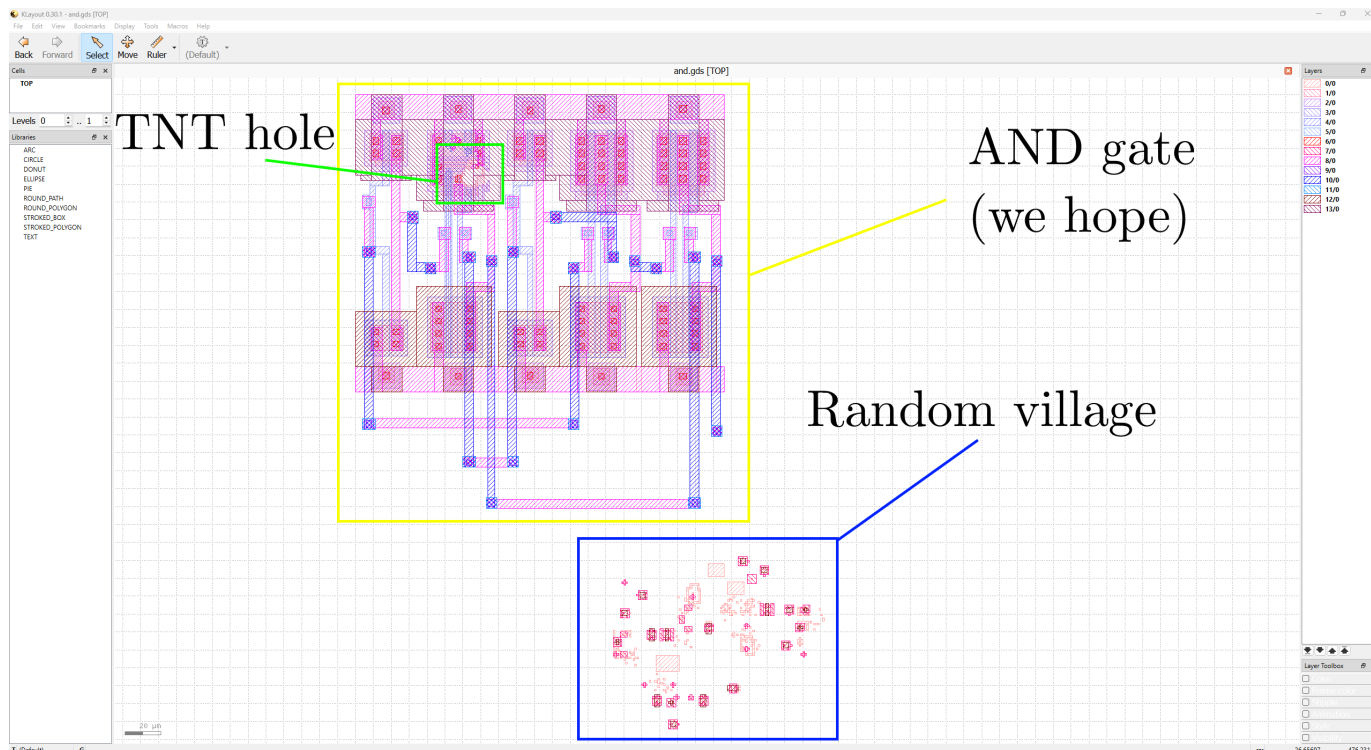


Fig. 3: Layout view of an AND gate after *editing* in MineGDS™. Note that some random village has appeared in the layout. The meaning behind the appearance of the village is to give a more realistic representation of what it would look like to live a life close to a giant floating circuit in the sky, hoping to not have your physical self, hopes, and dreams crushed.



Fig. 4: This is Steve, but we won't tell you which one.

Traders, Inc.) circulating, you will see that we created a way to always be profitable and never go out of business. This is 135% accurate, just trust us, also [11]. Our shareholders will always be the richest and our stock will never experience a downturn; simply impossible. Why hasn't anyone else figured this out yet? Who knows, maybe it's us providing the MOST, Inc. ®™© (*Monkeys Organizing Silly Tea-parties Inc.*) (get it?) value.

IV. CONCLUSION

MOST, Inc. ®™© (*Meteor Observation and Sandwich Testing, Inc.*) has presented MineGDS™, the world's first fully functional GDSII-editor in Minecraft [6]. Designs exported from MineGDS™ may (in theory) be manufacturable at numerous semiconductor foundries, around the world³.

³MOST, Inc. ®™© is not responsible for non-manufacturable designs created with MineGDS™

MineGDS™ enables a wide audience to adopt IC design, reinvigorating the field and guaranteeing a healthy semiconductor industry and world peace for decades to come.

You're welcome.

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MOST, Inc. ®™© (*Marketing Over Substance Technologies Inc.*) was founded in 1982 as the leading and trailblazing semiconductor concern in the world. At MOST, Inc. ®™©, we ignore Moore, soar beyond More-than-Moore and underscore nor abhor the chore to explore opening the door towards More-than-More-than-Moore. MOST, Inc. ®™© opens windows into the future of circuit design as the key synergetic building blocks of making technology that is the MOST technological.

